(i) Multiplication Theorem:

The probability of the simultaneous happening of two events A and B is given by

$P(A \cap B) = P(A).P(B/A)$ (or)

$P(A \cap B) = P(B).P(A/B)$

NOTE

If A and B are two independent events then $P(A \text{ and } B) = P(A \cap B) = P(A)\ P(B)$

The theorem can be extended to three or more independent events. Thus for three events the theorem states that

$P(A \text{ and } B \text{ and } C) = P(A \cap B \cap C)$

$= P(A)\ P(B) P(C)$

Example 8.24

An unbiased die is thrown. If A is the event 'the number appearing is a multiple of 3' and B be the event 'the number appearing is even' number then find whether A and B are independent?

Solution:

We know that the sample space is

$S = \{1,2,3,4,5,6\}$

Now, $A = \{3,6\}$; $B = \{2,4,6\}$ then $(A \cap B) = \{6\}$

$$P(A) = \frac{2}{6} = \frac{1}{3}$$

$$P(B) = \frac{3}{6} = \frac{1}{2} \text{ and } P(A \cap B) = \frac{1}{6}$$

Clearly $P(A \cap B) = P(A)\ P(B)$

Hence A and B are independent events.

Example 8.25

Let $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{5}$. Find $P(A \cap B)$ if A and B are independent events.

Solution:

Since A and B are independent events then

$P(A \cap B) = P(A)\ P(B)$

Given that $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{5}$,

then $P(A \cap B) = \frac{3}{5} \times \frac{1}{5} = \frac{3}{25}$

Example 8.26

Three coins are tossed simultaneously. Consider the events A 'three heads or three tails', B 'atleast two heads' and C 'at most two heads' of the pairs (A,B), (A,C) and (B,C), which are independent? Which are dependent?

Solution:

Here the sample space of the experiment is

S = {HHH, HHT, HTH, HTT, THH, TTH, THT, TTT}

A = {Three heads or Three tails}
= {HHH, TTT}

B = {at least two heads}
= {HHH, HHT, HTH, THH} and

C = {at most two heads} = {HHT, HTH, HTT, THH, TTH, THT, TTT}

Also $(A \cap B) = \{HHH\}$; $(A \cap C) = \{TTT\}$ and $(B \cap C) = \{HHT, HTH, THH\}$

$\therefore P(A) = \frac{2}{8} = \frac{1}{4}$; $P(B) = \frac{1}{2}$; $P(C) = \frac{7}{8}$ and

$P(A \cap B) = \frac{1}{8}$, $P(A \cap C) = \frac{1}{8}$, $P(B \cap C) = \frac{3}{8}$

Also $P(A).P(B) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$

$P(A).P(C) = \frac{1}{4} \cdot \frac{7}{8} = \frac{7}{32}$

and $P(B).P(C) = \frac{1}{2} \cdot \frac{7}{8} = \frac{7}{16}$

Thus, $P(A \cap B) = P(A).P(B)$

$P(A \cap C) \neq P(A).P(C)$ and

$P(B \cap C) \neq P(B).P(C)$

Hence, the events (A and B) are independent, and the events (A and C) and (B and C) are dependent.

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